

Practice Worksheet 2: Conditionals

CS 101 Algorithmic Problem Solving

Fall 2024

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1. To concert or not?

Four friends want to attend a concert and have a combined budget of Rs. 10,000. They will attend the concert only if the total cost is within the budget.

Given X , the cost of each ticket, determine whether they will attend the concert.

Constraints

- X
- $1 \leq X \leq 10000$

Interaction

The input contains a single line containing a single integer denoting the value of X .

The output must contain a single string containing “YES” if they will attend the concert, and “NO” otherwise.

Sample

Input	Output
1000	YES
5000	NO

In the first case, $X = 1000$. Four tickets would cost $1000 \times 4 = 4000$. The amount is within the budget and the friends will attend the concert.

In the second case, $X = 5000$. Four tickets would cost $5000 \times 4 = 20000$. The amount is not within the budget and the friends will not miss the concert.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2500	YES
3680	NO
8900	NO

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
1850	YES
4500	NO
780	YES

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: X

Output: if $X > 2500$: then print("NO"), else: print("yes")

2. Candies Fight

There are three friends and a total of N candies. There will be a fight amongst the friends if all of them do not get the same number of candies.

Chef wants to divide all the candies such that there is no fight. Find whether such distribution is possible.

Constraints

- $1 \leq N \leq 100$

Interaction

The input comprises a single line containing an integer denoting the value of N .

The output must be a Yes or No, depending on whether the candies can be evenly distributed or not.

Sample

Input	Output
3	Yes
4	No

In the first case, Chef can distribute all 3 candies such that each friend gets 1 candy. Since all three friends have same number of candies, there is no fight.

In the second case, There exist no way of distributing all candies such that all three friends have same number of candies.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2	NO
6	YES
123	YES

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
27	YES
89	NO
66	YES

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: N

Output: if $N \% 3 == 0$: print("YES"), else: print("no")

3. Gold Digger

On a vacation with N friends, Chef stumbled upon a gold mine and dug it all up. The total amount of gold is X kg. Every person has the capacity of carrying up *at most* Y kg of gold. Note that including Chef, there are a total of $(N + 1)$ people.

Given N , X and Y , will Chef and her friends be able to carry all the gold from the gold mine in a single go?

Constraints

- N, X, Y
- $1 \leq N, X, Y \leq 1000$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of N , X and Y respectively.

The output must contain a single string—"YES" if the gold can be carried in a single go, or "NO" if it cannot.

Sample

Input	Output
2 10 3	NO
2 10 4	YES

In the first case, $(N, X, Y) = (2, 10, 3)$. There are 3 people in total, each person can carry at most 3kg. The maximum amount of gold that can be carried is $3 \times 3 = 9$ kg, which is insufficient.

In the second case, $(N, X, Y) = (2, 10, 4)$. There are 3 people in total, each person can carry at most 4kg. The maximum amount of gold that can be carried is $3 \times 4 = 12$ kg, which is sufficient to carry all the gold in a single go.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4 25 6	YES
3 18 3	NO
1 9 4	NO

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
44 52 2	YES
76 34 6	YES
4 67 8	NO

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: X, Y, N

Output: if $(N+1)*Y \geq X$: then print("yes), else: print("NO")

4. Rooks Attack

You are given a standard 8×8 chessboard which has exactly 2 rooks on it and no other pieces. The rows are numbered 1 to 8 from bottom to top, and the columns are numbered 1 to 8 from left to right. The cell at the intersection of the i -th column and j -th row is denoted (i, j) .

Given the initial positions of the rooks in the form of coordinates (x_1, y_1) and (x_2, y_2) . you need to tell whether the 2 rooks currently attack each other or not. Assume, each square can contain at most one piece.

Hint: Rooks can only travel in straight lines along the row or column they are placed at, and can't jump over other pieces.

Constraints

- $1 \leq x_1, x_2, y_1, y_2 \leq 8$
- $(x_1, y_1) \neq (x_2, y_2)$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of x_1, y_1, x_2 and y_2 respectively.

The output must contain a single line stating Yes if the rooks attack each other, and No otherwise.

Sample

Input	Output
1 2 5 2	Yes
1 1 8 8	No

In the first case, The two rooks can attack each other by moving along the second column.

In the second case, No matter how a rook moves it cannot reach the second rook in one move. Hence, they do not attack each other.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1 2 1 5	YES
2 4 4 8	NO
3 3 4 3	YES

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
1 1 2 6	NO
2 4 4 2	NO
4 4 6 4	YES

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: x_1, x_2, y_1, y_2

Output: if $x_1 == x_2$ or $y_1 == y_2$: then print("yes"), else: print("NO")

5. Vaccination Dates

Chef took the first dose of vaccine D days ago. The second dose must be taken no less than L days and no more than R days after the first dose.

Given D, L and R , determine if Chef is too early, too late, or in the correct range for taking the second dose.

Constraints

- D, L, R
- $1 \leq D \leq 10^9$
- $1 \leq L \leq R \leq 10^9$

Interaction

The input contains a single line containing the space-separated integers indicating the values of D, L, R . respectively.

The output must contain a string—"Too Early" if it's too early to take the vaccine, "Too Late" if it's too late to take the vaccine, or "Take second dose now" if it's the correct time to take the vaccine.

Sample

Input	Output
8 8 12	Take second dose now
14 2 10	Too Late

In the first case, $(D, L, R) = (8, 8, 12)$. The second dose needs to be taken within 8 to 12 days. Day 8 lies in this range, Chef can take the second dose now.

In the second case, $(D, L, R) = (14, 2, 10)$. The second dose needs to be taken within 2 to 10 days. Day 14 lies after this range, it is too late now.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
4444 5555 6666	Too Early
8 10 12	Too Early
9 2 21	Take second dose now

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
34 45 88	Too Early
39 39 56	Take second dose now
77 45 99	Take second dose now

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: D, L, R

Output: if $D \geq L$ and $D \leq R$: print("Take second dose now"), elif $D < L$: print("Too early"), else: print("Too late")

6. Grading

HackerLand University has the following grading policy:

- Every student receives a grade in the inclusive range from 0 to 100.
- Any grade less than 40 is a failing grade.

Sam is a professor at the university and likes to round each student's grades according to these rules:

- If the difference between the grade and the next multiple of 5 is less than 3, round grade up to the next multiple of 5.
- If the value of grade is less than 38, no rounding occurs as the result will still be a failing grade.

Given the grade g of Sam's student, find out the final grade after rounding as per his rules.

Constraints

- $0 \leq g \leq 100$

Interaction

The input comprises a single line containing an integer denoting the value of g .

The output must contain a single number denoting final grade after rounding.

Sample

Input	Output
84	85
57	57

In the first case, $g = 84$ and the next multiple of 5 is 85, $85 - 84 = 1$ which is lesser than 3. Therefore, it is rounded to the next multiple of 5 which is 85.

In the second case, $g = 57$ and the next multiple of 5 is 60, $60 - 57 = 3$ which is not lesser than 3. Therefore, it is not rounded and remains as is.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
29	29
78	80
51	51

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
98	100
41	41
56	56

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: g

Output: if $g < 40$: then $\text{print}(g)$, else: $\text{rem} = g \% 5$, if $\text{rem} \geq 3$: $\text{print}(g)$, else: $\text{print}(g - \text{rem} + 5)$

7. Icebreaker

Chef has decided to try her skills as Karate. She is a part of a karate demonstration and her next task is to break three blocks of ice with widths W_1 , W_2 , and W_3 stacked on top of each other (W_1 is on the top and W_3 is on the bottom).

Chef's punch has a strength of S . Whenever she strikes at the stacked blocks, consider the largest $k \geq 0$ such that the sum of the width of the topmost k blocks does not exceed S ; the topmost k blocks shatter and are removed from the stack. Chef can also choose to reverse the order in which the blocks are stacked before every hit, without incurring any penalty.

Given values of W_1 , W_2 , W_3 , and S , find the minimum number of hits Chef needs to break through all three blocks if she performs the reversals optimally.

Constraints

- $W_1, W_2, W_3, S \in \mathbb{Z}$
- $1 \leq W_1, W_2, W_3 \leq S$
- $1 \leq S \leq 10$

Interaction

The input comprises a single line containing 4 space-separated integers denoting the value of W_1 , W_2 , W_3 , and S .

The output must be a single integer denoting the minimum number of hits required to break all three blocks if the reversals are performed optimally.

Sample

Input	Output
1 2 2 3	2
1 1 1 3	1

In the first case, $(W_1, W_2, W_3, S) = (1, 2, 2, 3)$. Chef can break through the first two blocks in one hit as $W_1 + W_2 \leq S$ and then the last block in a subsequent hit. Alternatively, she could flip the order, break through W_3 and then break through the remaining two blocks in the next hit. Minimum number of hits required is 2.

In the second case, $(W_1, W_2, W_3, S) = (1, 2, 2, 3)$. Chef can break through the three blocks in one hit as $W_1 + W_2 + W_3 \leq S$. Minimum number of hits required is 3

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
5 1 5 6	2
8 5 4 10	2
4 5 6 9	2

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
3 1 5 5	2
4 4 3 4	3
3 3 3 4	3

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: W1, W2, W3, S

Output: if $W1+W2+W3 \leq S$: then print("1"), elif $W1+W2 \leq S$ or $W2+W3 \leq S$: print("2"), else: print("3")

8. Preferential Treatment

Chef is hosting a small party at her house and has baked N cookies for her six guests. Unfortunately, she didn't put much thought into what amount of cookies she should bake, while all her guests seem to be really keen on trying them. Chef prefers some guests over others and has thought of this cookie division opportunity to be the perfect time to show her preference by giving a different amount of cookies to each guest depending on how much she likes them.

Chef wants to divide these N cookies among her guests such that:

- No two guests get the same amount of cookies.
- Each friend gets a cookie.

Given the value of N , determine if such a division is possible.

Constraints

- $N \in \mathbb{Z}$
- $1 \leq N \leq 10^5$

Interaction

The input comprises a single line containing a single integer denoting the value of N .

The output must be "YES" if a division according to the aforementioned method is possible and "NO" otherwise.

Sample

Input	Output
4	NO
42	YES

In the first case, $N = 4$. It is impossible to divide 4 cookies such that each of the six guests gets one.

In the second case, $N = 42$. A division according to the aforementioned method is possible. One solution could be 2, 4, 6, 8, 10, 12.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
20	NO
21	YES
10^3	YES

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
30	Yes
129	Yes
18	No

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: N

Output: If $N \geq 21$: then print("yes"), else: print("NO")

9. Who's better?

In a season of cricket, three metrics regarding a player are measured which are used to determine the quality of a player; these are runs (R), wickets (W), and catches (C). The player who is better in the most statistics is the better player. Below you are given the stats of two players A and B . The stats of player A are denoted as R_A, W_A, C_A and that of player B are denoted as R_B, W_B, C_B . Determine which is the better player.

Constraints

- $R_A, W_A, C_A, R_B, W_B, C_B \in \mathbb{Z}$
- $0 \leq R_A, R_B \leq 500$
- $0 \leq W_A, W_B, C_A, C_B \leq 20$

Interaction

The first line of the input contains 3 space-separated integers denoting the values of R_A, W_A , and C_A respectively. The second line of the input contains 3 space-separated integers denoting the values of R_B, W_B , and C_B respectively.

The output must be "A" or "B" depending on which player is better. In case of a tie, output "TIE".

Sample

Input	Output
0 1 2	"B"
2 3 4	
10 0 10	"A"
0 10 0	

In the first case, player "B" is better than player "A" in all three fields.

In the second case, player "A" is better than player "B" in two out of the three fields.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
34 5 6	
23 5 6	"A"
123 14 5	
323 1 6	"B"
199 14 16	
199 13 17	"TIE"

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
435 5 5	
56 8 9	"B"
49 3 2	
34 2 1	"A"
82 4 8	
85 3 8	"TIE"

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Input: Ra, Wa, Ca, Rb, Wb, Cb

Output: A=0, B=0, if Ra>Rb: then A=A+1, elif Rb>Ra: then B=B+1, else: break
 if Wa>Wb: then A=A+1, elif Wb>Wa: then B=B+1, else: break
 if Ca>Cb: then A=A+1, elif Cb>Ca: then B=B+1, else: break
 if A>B: print("A"), elif B>A: print("B"), else print("TIE")

10. Bouquet of Leaves

Chef wants to make a bouquet of leaves. It's fall season, so there's plenty of leaves on the ground. At her local park, she can find fallen leaves of maple, oak, and poplar trees. These leaves can be of three different colours: green, yellow, or red.

Chef has gone around the park picking up any leaf she sees and putting them in her basket. She wants to make a nice bouquet; a nice bouquet is one if and only if all the leaves in it are of the same type or colour. She also don't want to make a bouquet with an even number of leaves as she considers it bad luck. She lays out her collection of leaves and begins counting leaves of each number and type.

Given the number of leaves of each type, determine what is the maximum number of leaves she can have in a bouquet which satisfies the above requirements. She need not use all the leaves of a certain type/colour; e.g. if there are 20 red leaves, she can choose to use just 19 for the bouquet.

Constraints

- Number of leaves of each type/colour is a non-negative integer.

Interaction

The first line of the input contains 3 space-separated integers M_G , M_Y , and M_R denoting the values of a number of green, yellow, and red maple leaves respectively. The second line of the input contains 3 space-separated integers O_G , O_Y , and O_R denoting the values of a number of green, yellow, and red oak leaves respectively. The third line of the input contains 3 space-separated integers P_G , P_Y , and P_R denoting the values of a number of green, yellow, and red poplar leaves respectively.

The output must be a single integer denoting the maximum number of leaves Chef can have in a nice bouquet. The output must be 0 if it is impossible to create a nice bouquet.

Sample

Input	Output
1 2 3 3 2 1 1 3 4	7
3 5 13 25 4 1 2 19 0	29

In the first case, the sum of maple leaves is 6, the sum of oak leaves is 6, the sum of poplar leaves is 8, the sum of green leaves is 5, the sum of yellow leaves is 7, and the sum of red leaves is 8. The maximum number of leaves that can make a nice bouquet is 7.

In the second case, the sum of maple leaves is 21, the sum of oak leaves is 30, the sum of poplar leaves is 21, the sum of green leaves is 30, the sum of yellow leaves is 28, and the sum of red leaves is 14. The maximum number of leaves that can make a nice bouquet is 29.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
11 23 13 31 21 14 13 32 42	87
50 20 10 10 60 30 24 12 36	99
11 2 3 7 8 13 15 13 12	39

Propose

Provide sample inputs and outputs below. Do not reuse any of the values from above.

Input	Output
5 7 22 45 56 11 9 45 0	111
45 66 23 0 0 0 34 2 69	133
56 0 34 55 23 1 12 56 3	123

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

input: Mg, My, Mr, Og, Oy, Or, Pg, Py, Pr

Output: a=Mg+My+Mr

b=Og+Oy+Or
c=Pg+Py+Pr

d=Mg+Og+Pg
e=My+Oy+Py
f=Mr+Or+Pr

list=[a,b,c,d,e,f]
answer=max(list)
if answer%2==0:
 answer=answer-1
 print(answer)
else:
 print(answer)